

Trigonometric Identities and Integrals for Chapter 7

Math 132

Trigonometric Identities

$$\csc x = \frac{1}{\sin x} \qquad \sec x = \frac{1}{\cos x} \qquad \cot x = \frac{1}{\tan x}$$

$$\tan x = \frac{\sin x}{\cos x} \qquad \cot x = \frac{\cos x}{\sin x}$$

$$\sin^2 x + \cos^2 x = 1 \qquad \tan^2 x + 1 = \sec^2 x \qquad 1 + \cot^2 x = \csc^2 x$$

$$\sin x \cos x = \frac{1}{2} \sin 2x \qquad \cos 2x = \cos^2 x - \sin^2 x$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x) \qquad \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

Trigonometric Integrals and Non-Trigonometric Integrals

$$\int \sin x \, dx = -\cos x + C \qquad \int \cos x \, dx = \sin x + C$$

$$\int \sec^2 x \, dx = \tan x + C \qquad \int \csc^2 x \, dx = -\cot x + C$$

$$\int \sec x \tan x \, dx = \sec x + C \qquad \int \csc x \cot x \, dx = -\csc x + C$$

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C \qquad \int \csc x \, dx = \ln |\csc x - \cot x| + C$$

$$\int \tan x \, dx = \ln |\sec x| + C \qquad \int \cot x \, dx = \ln |\sin x| + C$$

Method for Evaluating Trig Integrals

ODD COSINE

- 1.) Factor out one power of cosine.
- 2.) Replace the remaining even powers of cosine using

$$\cos^2 x = 1 - \sin^2 x$$

- 3.) Let $u = \sin x$ and integrate.

ODD SINE

- 1.) Factor out one power of sine.
- 2.) Replace the remaining even powers of sine using

$$\sin^2 x = 1 - \cos^2 x$$

- 3.) Let $u = \cos x$ and integrate.

* If both sine and cosine are odd powered, you can use either of the above methods.

EVEN SINE AND COSINE

You will use the half angle identities.

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x) \quad \cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

Sometimes, you can use the following identity:

$$\sin x \cos x = \frac{1}{2} \sin 2x$$

EVEN SECANT

- 1.) Factor out a $\sec^2 x$.
- 2.) Replace the remaining even powers of secant using

$$\sec^2 x = 1 + \tan^2 x$$

- 3.) Let $u = \tan x$ and integrate.

ODD TANGENT

- 1.) Factor out a $\sec x \tan x$.
- 2.) Replace the remaining powers of tangent using

$$\tan^2 x = \sec^2 x - 1$$

- 3.) Let $u = \sec x$ and integrate.

Method for Trigonometric Substitution

Patterns	Substitute	Identity
$\sqrt{a^2 - x^2}$	$x = a \sin \theta$	$1 - \sin^2 \theta = \cos^2 \theta$
$\sqrt{a^2 + x^2}$	$x = a \tan \theta$	$1 + \tan^2 \theta = \sec^2 \theta$
$\sqrt{x^2 - a^2}$	$x = a \sec \theta$	$\sec^2 \theta - 1 = \tan^2 \theta$